

Patterns in the phonology of Mandarin Chinese: basic color terms and beyond

Daniel Hole and Titus von der Malsburg

University of Stuttgart

March 24, 2026

1. Two observations about basic color terms in Chinese

The six most important Chinese basic color terms (Berlin & Kay 1969) conform to a phonological system that, to the best of our knowledge, has not been described before. Consider (1) through (3).

(1) non-colors

- a. *bái* ‘white’
- b. *hēi* ‘black’

(2) warm colors

- a. *huáng* ‘yellow’
- b. *hóng* ‘red’

(3) cold colors

- a. *lán* ‘blue’
- b. *lǜ* ‘green’

Ignoring the rimes and only looking at the onsets of the non-colors in (1) for a start, one observes a rather large phonological/phonetic difference which corresponds to the large perceptive contrast between black and white. Expressed in articulatory features, *bái* ‘white’ starts with a voiced or non-aspirated front (bilabial) plosive, whereas *hēi* ‘black’ starts with a voiceless back (uvular or glottal) fricative (there is free variation between uvula/velum and glottis for place of articulation in Chinese with /h/; Duanmu 2007: 25). Hence, the plosive unaspirated or voiced manner of articulation in *bái* ‘white’ forms a stark contrast with the voiceless fricative manner of articulation in *hēi* ‘black’. Moreover, the bilabial place of articulation constitutes the foremost of all possibilities, whereas the uvular/glottal place of articulation of *hēi* constitutes the backmost of all possibilities.

The terms for the basic warm colors yellow and red in (2) both have /h/ in the onset, thereby establishing a phonological/phonetic similarity that may reflect the shared class-membership of yellow and red in the set of warm colors. The terms for the basic cold colors

green and blue in (3) both have the alveolar lateral glide /l/ in the onset, again establishing a parallel phonological/phonetic similarity in the realm of the cold colors.

Turning to the rimes now, we note strong similarities among the non-color terms, and among the warm color terms. For the rest of this study, but with one exception in the immediately following paragraph, we will pay no further attention to these rime similarities, leaving their closer scrutiny for a future occasion. We would like to make it clear, though, that the integration of these observations into our experiments would certainly yield even more robust results.

One has to know in connection with all this that Chinese is a language with a heavily restricted syllable structure. Accordingly, the number of possible syllables, abstracting away from tone and dialectal /r/-suffixed variants, amount to a set with just 403 members. (This number is an approximation. There exist syllables that are clear contractions of particles like *lo*, which is commonly analyzed as *le+o*, and there are syllables of a clearly ideophonic nature of the non-abstract kind such as *diang* ‘the sound of bells’. We disregard syllables like these.) Subtracting the syllables with no (phonological) onset, we arrive at 362 syllables.

Note in passing that the observed correspondences are not those that are found among people with synesthesia (colored hearing). The phoneme /h/, for instance, if realized as the voiceless uvular fricative [χ], will often be associated with the color of coffee (Jakobson 1941) or grey (Natalie Bour, p.c.) in humans with synesthesia. Neither of these two colors is a typical warm color, but in Chinese /h/ appears to single out the warm colors. Likewise, /a/ almost uniformly maps to red for people with this gift (Jakobson 1941, Nathalie Bour p.c.), but /a/ is not a phoneme in *hóng* ‘red’. Therefore, the basis of the system as found in Chinese cannot be one that is motivated by colored hearing/synesthesia.

What we are dealing with here is, we submit, some very abstract kind of ideophony, the phenomenon of portions of the lexicons of individual languages being motivated instead of arbitrary in their phonetic appearance (cf. Dingemanse 2012 and the works cited there). “Abstract” in the sense that, for instance, nothing in /h/ makes one associate the realization of this phoneme with a non-color; instead, identical onset phonemes, whatever they are, constitute class membership in well-established sensory classes (non-colors, cold and warm colors).

In the experiment to be reported on in section 2, we capitalize on the hypotheses in (4).

- (4) a. It is not by accident that the onsets of the basic color terms of Chinese single out the well-established subkinds of colors (non-colors, cold and warm colors).

- b. It is not by accident that the non-color terms feature onsets that differ in terms of manner of articulation, and differ maximally in terms of place of articulation.

The competing null hypothesis would say that none of the correlations in (4) hold and that the basic color terms of Chinese draw their onsets from the available onset types in a random fashion.

2. Null hypothesis testing

To rule out the possibility that the observed patterns are merely random coincidences, we conducted simulations of the expected onset patterns assuming that there are no systematic relationships of the kinds described above (null hypothesis for (4a) and (4b), respectively). We then calculated how likely it is that the observed patterns predicted by (4a) and (4b) occur by chance. If this probability is below the conventional α -level of 0.05, we conclude that the observed pattern occurred unlikely by chance.

To test (4a) we ran a computational simulation with 100,000 iterations. In each iteration, we sampled four random onsets from the full set of 21 which were weighted by their probability of occurrence, i.e. the proportion of Chinese syllables (types) that start with that onset. These 4 onsets were randomly assigned to the four colors in (2)–(3). We then counted cases in which yellow and red had the same onset, and green and blue had the same onset but one that was different from yellow/red. This pattern occurred in 0.2% of the iterations ($p=0.002$), which suggests that the observed pattern unlikely occurred by chance.

To test (4b) we conducted a similar simulation. In each iteration, we sampled two random onsets which were randomly assigned to black and white. We then counted cases in which the onsets of black and white formed a maximal contrast between anterior and posterior onsets which also differed in their manner of articulation. Such patterns occurred in 3% of the cases, which again suggests that the pattern was unlikely a random coincidence. The probability that the patterns predicted by (4a) and (4b) occur simultaneously by chance was $p<0.0001$, i.e. this is expected to occur in only 1 out of 15,522 cases. The R code for this simulation can be found in Appendix A.

3. Beyond colors

It seems safe to say that the observed patterns of onset distribution in Chinese basic color terms are not random coincidences. We would like to widen the perspective now and take a cursory look at some very common adjectival antonym pairs.

- | | | | |
|--------|-----------------------|--------------------------|-----------------------------------|
| (5) a. | <i>dà : xiǎo</i> | ‘big : small’ | [alveolar vs. alveolo-palatal] |
| b. | <i>gāo : āi</i> | ‘tall : short’ | [velar vs. ʃ] |
| c. | <i>zhòng : qīng</i> | ‘heavy : light’ | [retroflex vs. alveolo-palatal] |
| d. | <i>rè/nuǎn : lěng</i> | ‘hot/warm : cold’ | [retroflex/alveolar vs. alveolar] |
| e. | <i>cháng : duǎn</i> | ‘long : short’ | [retroflex vs. alveolar] |
| f. | <i>cū : xì</i> | ‘thick : thin’ | [alveolar vs alveolo-palatal] |
| g. | <i>nán : róngyì</i> | ‘difficult : easy’ | [alveolar vs. retroflex] |
| h. | <i>duō : shǎo</i> | ‘much/many : little/few’ | [alveolar vs. retroflex] |

Almost all onsets belong to the alveolars, the alveolo-palatals, or the retroflex sounds. Front (bilabial/labiodental) and back (velar/uvular/laryngeal) sounds are missing (with the exception of *gao* ‘tall’ in (5b)). At the same time, all but one pair (*nuan* : *leng* ‘warm : cold’ in (5d)) do exploit the contrasts offered by the three observed places of articulation common with antonym pair onsets.

It would seem to us that the study of well-structured lexical fields of Mandarin Chinese may deserve closer scrutiny to unravel further patterns like these.

References:

- Berlin, B., & Kay, P. (1969). *Basic Color Terms: their universality and evolution*. Berkeley: University of California Press.
- Dingemanse, M. (2012). Advances in the cross-linguistic study of ideophones. *Language and Linguistics Compass* 6(10), 654-672.
- Duanmu, S. (2007). *The Phonology of Standard Chinese*. Oxford: Oxford University Press.
- Jakobson, R. (1941). *Kindersprache, Aphasie und allgemeine Lautgesetze*. Uppsala: Almqvist & Wiksell.

Appendix A: Simulation code

```
library(dplyr)
options(scipen=999)

N <- 100000 # Number of iterations

# Data:

onsets <- tibble(
  onset = c("b", "p", "m", "f", "d", "t", "n", "l", "g", "k", "h",
            "j", "q", "x", "zh", "ch", "sh", "r", "z", "c", "s"),
  count = c(16, 16, 18, 9, 21, 19, 23, 24, 18, 18, 19, 13, 14, 14, 20,
            18, 19, 14, 17, 16, 16),
  prob = count / sum(count))

# Assuming that color terms have random onsets
# (H0), how likely is it that warm colors (yellow,
# red) share the same onset and cold colors (blue,
# green) share the same onset?

sapply(1:N, function(i) {
  with(onsets,
    sample(onset, 4, replace=TRUE, prob=prob))
}) |>
  t() |>
  data.frame() |>
  dplyr::rename(yellow=X1, red=X2, blue=X3, green=X4) -> sim

# Count cases where "yellow" and "red", and "blue"
# and "green" share the same onset:

sim |>
  filter(
    yellow == red,
    blue == green,
    yellow != green) |>
  nrow() -> i

# p-value:
i/N -> p1

# Assuming that "black" and "white" have random
# onsets (H0), how likely is it that they start
# with onsets that are extremely dissimilar with
# respect to place and mode of articulation?

# Assign random onsets to black and white and
# simulate N time:

sapply(1:N, function(i) {
```

```

with(onsets,
      sample(onset, 2, replace=TRUE, prob=prob))
}) |>
t() |>
data.frame() |>
dplyr::rename(black=X1, white=X2) -> sim

# Count cases with extremely dissimilar onsets:
sim |>
  filter(
    case_when(
      black %in% c("p", "b") & white %in% c("h") ~ TRUE,
      black %in% c("m") & white %in% c("g", "k", "h") ~ TRUE,
      black %in% c("g", "k") & white %in% c("m", "f") ~ TRUE,
      black %in% c("f") & white %in% c("g", "k") ~ TRUE,
      white %in% c("p", "b") & black %in% c("h") ~ TRUE,
      white %in% c("m") & black %in% c("g", "k", "h") ~ TRUE,
      white %in% c("g", "k") & black %in% c("m", "f") ~ TRUE,
      white %in% c("f") & black %in% c("g", "k") ~ TRUE,
      .default = FALSE)
  ) |>
  nrow() -> i

# p-value:
i/N -> p2

p1 * p2 -> p3

message("p1: ", p1)
message("p2: ", p2)
message("p3: ", p3)

```